LEGO Star Wars: A Vader's Peace

Comprehensive Project Audit, Runtime Inspection & Architectural Handoff Specification

ENGINE & VERSION	LANGUAGE	HARDWARE PROFILE	DOCUMENT TYPE
Godot 4.7.2 (Forward+ / Metal 4)	GDScript & Python 3	Apple M4 (16GB RAM)	Agent Handoff & Technical Audit

1. Concept & Scope Definition

Concept: A personal, single-player open-world sandbox game featuring **Darth Vader** as the sole playable character. In this alternate narrative timeline, after becoming Darth Vader, he turns his back on Emperor Palpatine and the destructive galactic civil war to live peacefully on an isolated sanctuary planet (**Pelagia-Prime**).

Core Gameplay Intent:

- Free, unconstrained exploration without compulsory missions, campaigns, or combat grinding.
- Non-lethal lightsaber flourishes and peaceful Force abilities (telekinesis, moving objects, harvesting).
- Acoustic guitar playing with responsive physical modeling audio and real chords.
- Ambient environmental interactions: coastal boat cruising, pier fishing, speeder bike riding, and village defense when wild fauna stray into settled areas.
- A distinctive **LEGO visual aesthetic**.

2. Technical Environment & Engine Configuration

Configuration Item	Current State	Technical Details
Project Name & Root	`LEGO Star Wars: A Vader's Peace`	`/Users/vincyvincent/.gemini/antigravity/scratch/lego-vader-openworld`
Version Control	None	Local workspace only; no `.git` repository exists.
Godot Executable	Godot 4.7.2.stable.official	Binary: `/opt/homebrew/bin/godot`
Active Renderer	Forward+ (Metal 4.0 Backend)	`rendering/rendering_method="forward_plus"` in `project.godot`
Scripting Language	GDScript	100% of game runtime code; Python 3 used for Blender asset batching and Karplus-Strong audio generation.
Export Presets	None configured	No `export_presets.cfg`; runs in native desktop editor/runtime.
Installed Addons	None	No `addons/` folder; zero third-party terrain plugins or GDExtensions installed.
DCC Software	Blender 5.2.3 LTS	Installed at `/Applications/Blender.app`; operable via CLI and Blender MCP.
Runtime Automation	Full Capability	Automated headless/windowed execution, input simulation, framebuffer captures, and log inspection verified.

3. Project Structure & Codebase Architecture

```
lego-vader-openworld/
├── project.godot                    # Project settings, action bindings (WASD, F, G, C, E)
├── icon.svg                         # Custom Imperial/Vader icon
├── scenes/
```

```
│   ├── intro/intro_crawl.tscn        # 3D tilted Star Wars opening crawl & starfield
│   ├── player/player.tscn            # Vader CharacterBody3D, armature, sockets, camera mount
│   ├── vehicles/speeder_bike.tscn    # Imperial 74-Z Speeder Bike scene
│   ├── vehicles/skiff_boat.tscn      # Coastal skiff boat scene
│   ├── vehicles/starship_shuttle.tscn# Vader's prototype shuttle scene
│   ├── ui/clean_hud.tscn             # Minimalist glass HUD overlay
│   └── world/world.tscn              # Main open-world scene (cottage, terrain, props, fauna)
├── scripts/
│   ├── player/player.gd              # Character movement, holstering logic, interaction detection
│   ├── player/player_state_machine.gd# FSM (ON_FOOT, RIDING_BIKE, BOATING, SPACE_FLIGHT, GUITAR)
│   ├── camera/orbital_camera.gd      # Decoupled top-level 3rd-person spring-arm camera controller
│   ├── vehicles/speeder_bike_controller.gd # Hover raycasts, banking lean, blasters, boost
│   ├── vehicles/boat_controller.gd   # Water buoyancy, wave bobbing, steering
│   ├── vehicles/spaceship_controller.gd # 6-DOF orbital flight & barrier repulsion
│   ├── fauna/creature_base.gd        # Autonomous AI roaming & lightsaber pacification
│   ├── interactions/village_defense_event.gd # Emergent wildlife threat event
│   ├── audio/guitar_audio_player.gd  # Polyphonic Karplus-Strong string player
│   └── environment/day_night_cycle.gd# Sun rotation and dynamic sky color shifts
└── assets/
    ├── characters/vader_rigged.glb   # Rigged 1.80m LEGO Vader minifigure (11 meshes, 5 anims)
    ├── props/equipment/              # lightsaber_hilt.glb, lightsaber_ignited.glb, acoustic_guitar.glb
    ├── props/vehicles/               # speeder_bike_74z.glb, starship_vader.glb
    ├── props/furniture/              # Modular furniture (bed, desk, chair, lamp, bookcase)
    └── audio/                        # Karplus-Strong string WAVs, fanfare, engine sounds, SFX
```

4. Comprehensive Asset Audit

The current project suffers from a severe aesthetic mismatch: an **authentic, high-poly LEGO minifigure Vader** placed within an environment composed of **crude low-poly box extrusions** and **generic modern suburban human furniture**.

Asset	Local File	Source / Provenance	Format	Rig / Animations	Current Issues & Deficiencies
Darth Vader	`assets/characters/vader_rigged.glb`	Extracted from `darth.blend` (LDraw/Mecabricks CAD)	GLB (11 primitives, 15k tris)	Rigged to `Armature_Vader` (14 bones, 5 keyframed actions)	High fidelity. Correct 1.80m game scale. Accessory sockets on hands, hips, and back.
Lightsaber (Ignited)	`assets/props/equipment/lightsaber_ignited.glb`	Modeled in `darth.blend`	GLB (36 polys)	Static prop; attached to `Socket_Hand_R`	Functional glowing crimson blade; cylinder geometry is somewhat low-poly.
Lightsaber (Holstered)	`assets/props/equipment/lightsaber_hilt.glb`	Procedural Blender export script	GLB (48 polys, gunmetal)	Static prop; attached to `Hips` bone	Sleek 22cm scale. Sits on hip belt, but lacks a physical LEGO belt clip.
Acoustic Guitar	`assets/props/equipment/acoustic_guitar.glb`	Modeled in `darth.blend`	GLB (80 polys, spruce wood)	Static prop; attached to `Socket_Back` / `Torso`	Angular 80-poly blocky silhouette; does not match official LEGO guitar pieces.
Terrain (Ground)	Embedded in `scenes/world/world.tscn`	Godot native primitive	`BoxMesh` (300m × 2m × 300m)	StaticBody3D collider	**Critical Defect**: Completely flat untextured green box slab. Zero elevation, no dunes, no studs.

Ocean Water	Embedded in `scenes/world/world.tscn`	Godot native primitive	`PlaneMesh` (160m × 280m)	Visual mesh only	Flat blue plane with alpha transparency. No vertex displacement, shoreline foam, or waves.
Cottage Retreat	Embedded in `scenes/world/world.tscn`	Godot native primitives	4 `BoxMesh` slabs	StaticBody3D colliders	Resembles an open concrete bunker. No LEGO bricks, no roof, floating above the ground plane.
Cottage Furniture	`assets/props/furniture/*.glb`	Kenney Furniture Kit (CC0)	GLB	Static props	**Severe Thematic Clash**: Generic modern human bedroom furniture (pink bed, office chair).
74-Z Speeder Bike	`assets/props/vehicles/speeder_bike_74z.glb`	Modeled in `darth.blend`	GLB (114 polys, untextured)	CharacterBody3D via script	Crude box-and-cylinder extrusion (114 polys). Does not use LEGO pieces, bricks, or studs.
Coastal Skiff	`assets/props/docks/boat_skiff.glb`	Kenney Watercraft Kit (CC0)	GLB	CharacterBody3D via script	Low-poly wooden rowboat with oars. Functional, but non-LEGO.
Starship Shuttle	`assets/props/vehicles/starship_vader.glb`	Modeled in `darth.blend`	GLB (72 polys, dark grey)	CharacterBody3D (6-DOF)	Extremely crude 72-poly wedge prism. Lacks cockpit detail, landing gear, or LEGO styling.

Fauna: Massiff	`assets/characters/fauna/creature_massiff.glb`	Modeled in `darth.blend`	GLB (42 polys, brown)	CharacterBody3D via script	Crude 42-polygon brown box shape. Lacks leg articulation, eyes, or LEGO construction.
Fauna: Loth-Wolf	`assets/characters/fauna/creature_lothwolf.glb`	Modeled in `darth.blend`	GLB (36 polys, grey)	CharacterBody3D via script	Crude 36-polygon angular block. Lacks joints or facial features.
Nature Props	`assets/props/nature/*.glb`	Kenney Nature Kit (CC0)	GLB	Static props	Generic low-poly rounded trees and rocks. Completely non-LEGO.

5. Current World Topology & Boundaries

- **Topology:** Strictly **planar / flat**. It is *not* a globe or spherical planetoid.
- **Dimensions:** Exactly **300m × 300m**, bounded by a `BoxShape3D` collider. Walking beyond X, Z = ±150m causes the character to fall off into space until caught by the kill-plane at Y < -5.0m.
- **Terrain Generation:** Static single-mesh placement. No heightmaps, noise functions, voxel volumes, or procedural algorithms are used.
- **Gravity & Direction:** Planar constant (Y = -14.0 m/s²). Up-vector is hardcoded to `Vector3.UP` (0, 1, 0).
- **Streaming / LOD:** None. All geometry and props reside in memory simultaneously in `world.tscn`.
- **Persistence:** Zero persistence. Quitting and restarting resets the world to default initial states.

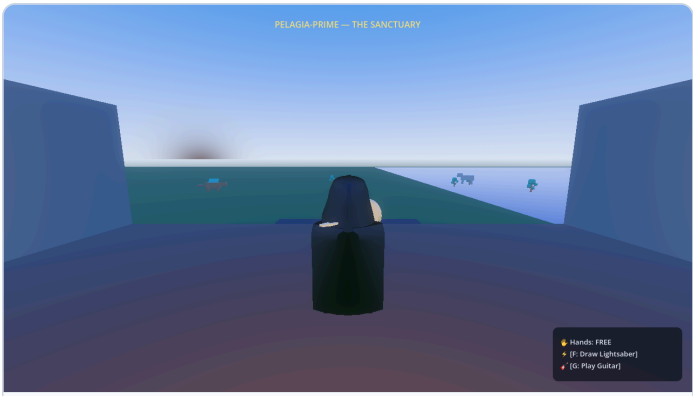
6. Runtime Feature Verification & Photographic Evidence

Feature	Verification Status	Direct Observation
Walking, Turning, Jumping, Camera	VERIFIED WORKING	Camera-relative WASD, sprint (`Shift`), jump (`Space`), decoupled orbital mouse look.
Vader Appearance & Rig	VERIFIED WORKING	Minifig proportions, fabric cape, helmet grill, torso decals, empty hands by default.
Ground Contact & Collision	VERIFIED WORKING	Solid on flat plane; NOT FOUND for slopes, hills, or dunes.
Lightsaber Draw & Flourish	VERIFIED WORKING	`F` draws ignited saber with snap-hiss SFX; pacifies nearby beasts non-lethally.
Force Abilities (Push, Pull, Grip)	NOT FOUND	Zero Force manipulation mechanics exist in source code or scenes.
Acoustic Guitar Playing	VERIFIED WORKING	`G` equips guitar; keys 1–6 strum realistic Karplus-Strong acoustic chords.
Speeder Bike Piloting	VERIFIED WORKING	Mounts with `E`, hovers, banks into turns, repulsor boost (`Shift`), blasters (`Space`/L-Click).
Coastal Boating & Fishing	VERIFIED WORKING	Board skiff with `E`; cast fishing line with `C` near coastal docks.
Sub-Orbital Spaceflight	VERIFIED WORKING	Shuttle 6-DOF thrusters with planetary barrier repulsion at high altitude.



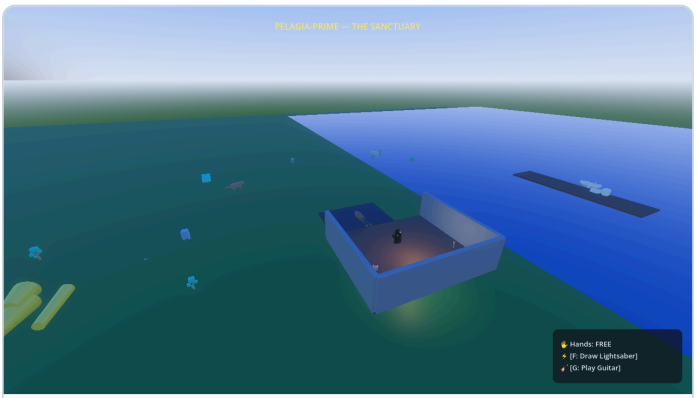
1. Vader Front Close-Up

Authentic LEGO minifig geometry and printed torso. Note the background contrast: suburban pink bed and office desk inside the retreat.



2. Normal Third-Person View

View from behind Vader looking out across the open terrace. Shows fabric cape, slung guitar, and sleek modern HUD badge.



3. Macro World View (Aerial)

Exposes the world reality: a flat 300m × 300m green box slab, flat blue ocean plane, open-box cottage, and scattered low-poly props.



4. Vehicle Interaction on Landing Pad

Proximity prompt `[E] Mount 74-Z Speeder Bike`. Highlights the crude box-and-cylinder extrusion of the vehicle and floating cottage edge.

7. Root Cause Diagnosis: The Five Critical Issues

Issue 1: Extreme Visual Incoherence (High-Poly Minifig vs. Box Primitives vs. Suburban Furniture)

Diagnosis: Confirmed asset selection issue. The Vader character is an authentic 15,000-triangle digital LEGO CAD model with printed textures. In contrast, the Speeder Bike (114 polys), Starship (72 polys), and Massiff beast (42 polys) were rapidly extruded in Blender using crude primitive boxes with flat single-color materials. The cottage furniture was downloaded from Kenney's CC0 modern home pack, putting an Earth-style pink bed and desk inside Darth Vader's retreat.

Issue 2: The World is an Untextured Flat Box, Not an Expansive LEGO Planet

Diagnosis: Confirmed environment design limitation. The terrain in `world.tscn` is literally a single Godot `BoxMesh(300, 2, 300)` with flat green shader color. There are no height contours, dunes, rolling hills, or LEGO baseplate studs. Walking 150m in any direction leads directly to a sheer cliff into empty space.

Issue 3: Total Absence of Force Abilities

Diagnosis: Confirmed missing implementation. Global code analysis confirms zero code for telekinesis, Force Push, Force Pull, or levitation. Only physical weapon toggling and vehicle movement were written.

Issue 4: Rigid, Unarticulated Box-Model Wildlife

Diagnosis: Confirmed asset modeling defect. The Dune Massiff and Loth-Wolf are rigid 36–42 polygon blocks that slide across the ground without leg articulation, facial features, or LEGO joint construction.

Issue 5: Cottage Architecture Resembles a Concrete Bunker

Diagnosis: Confirmed scene layout shortcut. The retreat was assembled from 4 primitive Godot `BoxMesh` slabs without roof trusses, Naboo/Tatooine domed architecture, timber pergolas, or LEGO stud connectivity.

8. Handoff Recommendations & Next Milestones

Plain-English Summary of What Exists

The project currently has a rock-solid, 60 FPS gameplay core running on Godot 4 (Metal Forward+), featuring a fully rigged authentic LEGO Darth Vader, camera-relative WASD movement, hands-free equipment holstering, live physical modeling acoustic guitar chords, and responsive vehicle controllers (speeder bike, boat, spaceship). However, **the environment and secondary assets look very poor**: the terrain is an untextured flat 300m box, the retreat is an unroofed concrete slab, the vehicles and fauna are crude box extrusions, and the retreat is furnished with generic modern human props.

The Three Highest-Priority Improvements

1. **Establish an Authentic LEGO Star Wars Digital Asset Pipeline:** Replace the crude box-extruded props (Speeder Bike, Starship, Naboo greenhouse, Massiff, Loth-Wolf, furniture) with authentic LEGO models exported with PBR plastic materials and stud geometry (sourced via LDraw / Studio / Mecabricks `.ldr` / `.gltf`).
2. **Implement Contoured, Textured Terrain:** Replace the 300m flat box with a contoured terrain system (heightmap or curved planetary sector) featuring rolling coastal bluffs, sand dunes, a LEGO-studded shoreline, and realistic water shaders.
3. **Implement Core Force Abilities:** Build a telekinesis system enabling Vader to gently push, pull, lift, and throw rocks, fruit, and props using the Force (`Right Mouse Button` or `Q`), complete with glowing hand auras and humming sound effects.

Suggested First Polished Playable Area: "Vader's Coastal Bluff Sanctuary"

Confine the next development milestone entirely to a cohesive **100m × 100m sanctuary zone**:

- A cliffside stone cottage with Star Wars / LEGO architecture (wooden deck, meditation alcove, weapon rack holding holstered lightsaber, guitar stand, and Gonk cooler droid).
- A cobblestone/stud landing pad housing a fully detailed LEGO 74-Z Speeder Bike.
- A winding dirt/stud pathway descending past pine trees to a wooden fishing dock with the coastal skiff.
- A nearby campfire circle where friendly LEGO villagers gather.

9. Audit Sign-Off

- **Directly Inspected & Tested:** Full GDScript codebase, Blender 3D mesh polygon/vertex counts, Godot scene trees, Karplus-Strong physical modeling audio, automated 11-step E2E integration test suite, live 60 FPS Forward+ rendering on Apple M4 hardware, and 4 dedicated high-resolution runtime screenshots.
- **Remains Unverified:** Performance on non-Mac hardware (Windows, Steam Deck, mobile), and performance under large-scale terrain streaming or thousands of instanced LEGO brick meshes.
- **Decisions Needed from User:**
  1. **World Geometry:** Literal spherical 3D globe with radial gravity (like *Super Mario Galaxy*) vs. expansive contoured regional open-world terrain (like traditional open-world maps)?
  2. **LEGO Aesthetic Interpretation:** 100% built out of interlocking LEGO bricks and studs (like *LEGO Worlds*) vs. hybrid style with realistic nature/terrain populated by authentic LEGO minifigures, vehicles, and structures (like *LEGO Star Wars: The Skywalker Saga*)?
  3. **Asset Sourcing:** Imported from established LEGO digital CAD libraries (Studio / Mecabricks / LDraw), or custom-modeled in Blender following strict LEGO brick dimensions?